MSc Extended Research Project EMS715P
MSc Final Report & Viva Lecture

Coursework completed to date include:

DLA1,
DLA2,
RAO Poster &
RAO Presentation.
SOTA

Dr Raza Shah

Final Report – Guide to Key Sections and Structure

Reports are written in several sections and may also include visual data such as figures and tables. The layout and presentation is therefore very important.

1. Signposting Sections, Abstract, Declaration & Acknowledgements

2. Introduction, Background and Literature Review

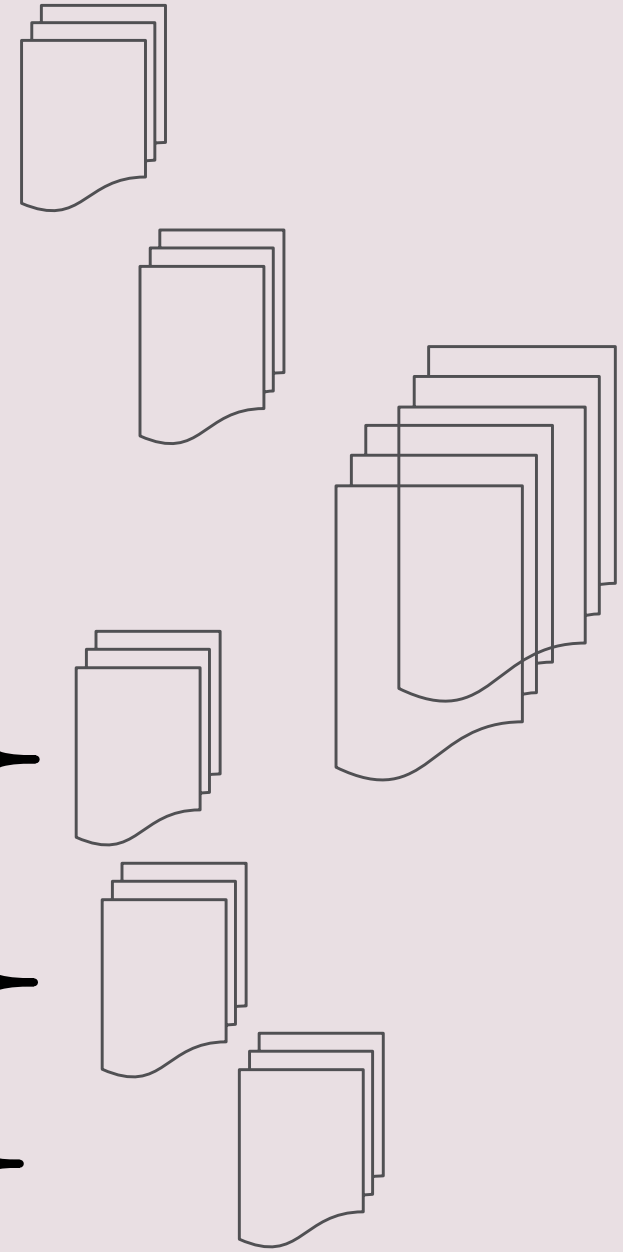
3. Report/Dissertation <u>Main Body</u>

- **What you did – how and why?**
- **Results and outcomes?**
- **Validity and reliability?**
- **Conclusions?**

4. Discussion

5. References

6. Appendices



<u>"Guide" to section lengths</u>

Introduction and Literature review
~1200

Methodology
~800

Results, analysis, findings and conclusions
~3500

Discussion
~500

Planning Activity:
Constructing Your Final Report Template
*See also the EMS715P Handbook

DO NOT USE QM LOGO

1. Initial Sections

Title page

Author declaration

Abstract (or Executive Summary) *One or two paragraph summary of the contents report. It should comprise a brief and factual account of the contents including a statement of major conclusions reached. Read by more people than any other part. It should state the methods or procedures followed, the principal findings, and the conclusions.*

Acknowledgements

Contents page

Nomenclature

2. Introduction

Background/Literature review – remember that the content of your literature review section should now be much more focused and fine-tuned to the project final RAO. Use the relevant format and referencing convention as agreed with you project supervisor and be consistent.

Report/Dissertation Main Body

3. Method, Methodology or Research Design - the main difference between research methods and research design is that research design is the overall structure of the research study whereas research methods are the various processes, procedures, and tools used to collect and analyze data.

5. Results/findings

6. Conclusions - a concise summary of appropriate concluding statements brought up in the discussion. You should NEVER introduce in this section any new material or ideas that have not been included in the main body**.**

7. Discussion – recommendations and future work

8. References - Everything you write is presumed to be your own work if it is not referenced. You must show the reader which parts are not your own work. If you use other people's work and do not reference it correctly you are plagiarising. Other "work" covers words, images, tables, charts, definitions, facts and figures.

Appendices – Only use the appendices for directing the reader to more detailed information relevant to the report. Do not just add them without signposting the reader in the main body of the report – place the appendices within the context of the report and do not rely on the reader to interpret your appendices without adequate sgnposting.

*There will naturally be some variation in the nature/methodology/methods across MSc projects so consult your project supervisor to check on the formatting, referencing conventions and structure.

Sections and Structure

1. Signposting Sections

*__Guide:__

DO NOT USE QMUL LOGO

Title page

Author declaration

Abstract t(or Executive Summary)

Acknowledgements

Contents page

Sections and Structure

2. Introduction, Background and Literature Review

Introduction
Background/Literature review

Sections and Structure

3. Report/Dissertation Main Body

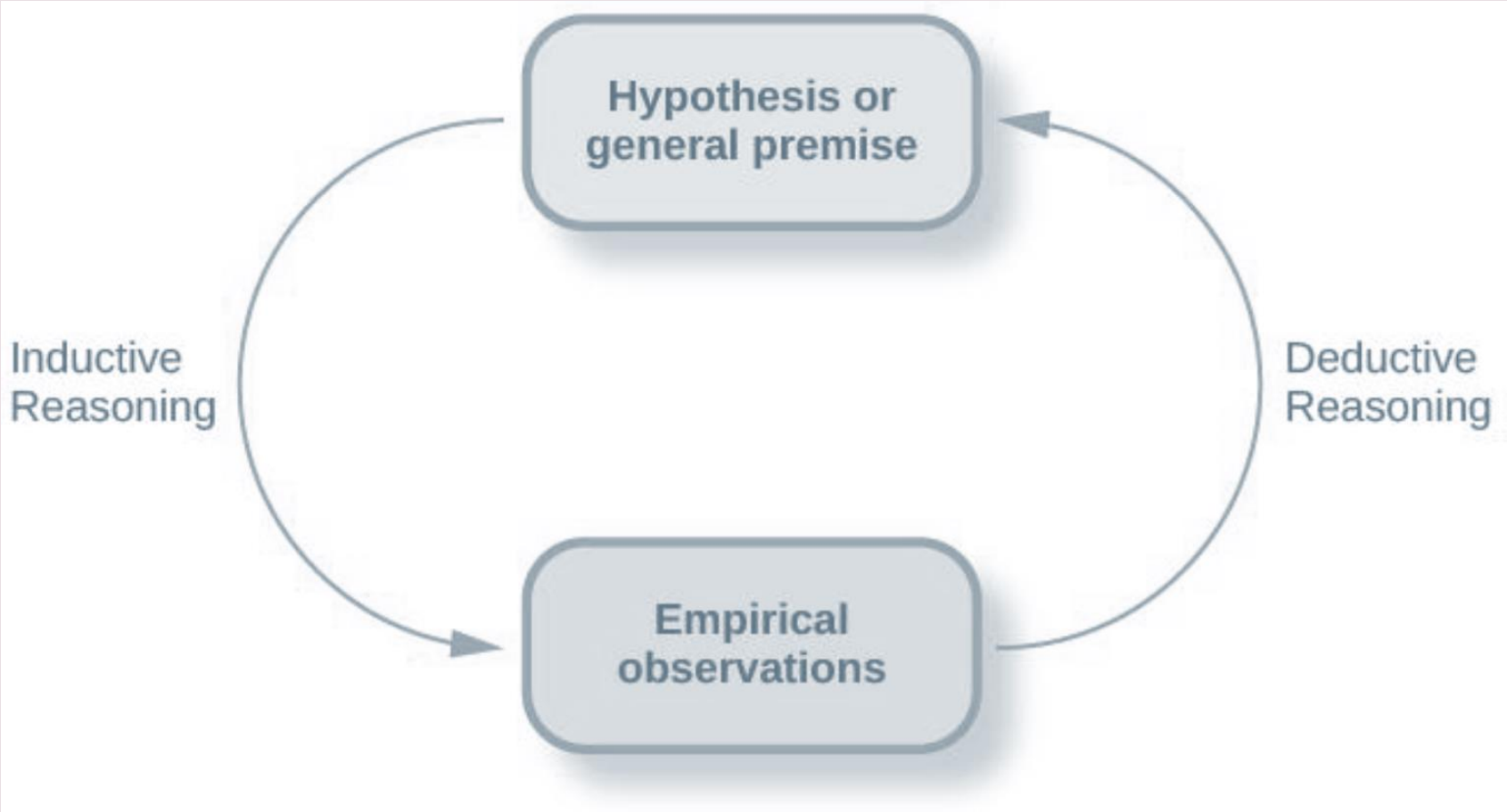
Method, Methodology or Research Design
Method, Methodology or Research Design - the main difference between research methods and research design is that research design is the overall structure of the research study whereas research methods are the various processes, procedures, and tools used to collect and analyze data.

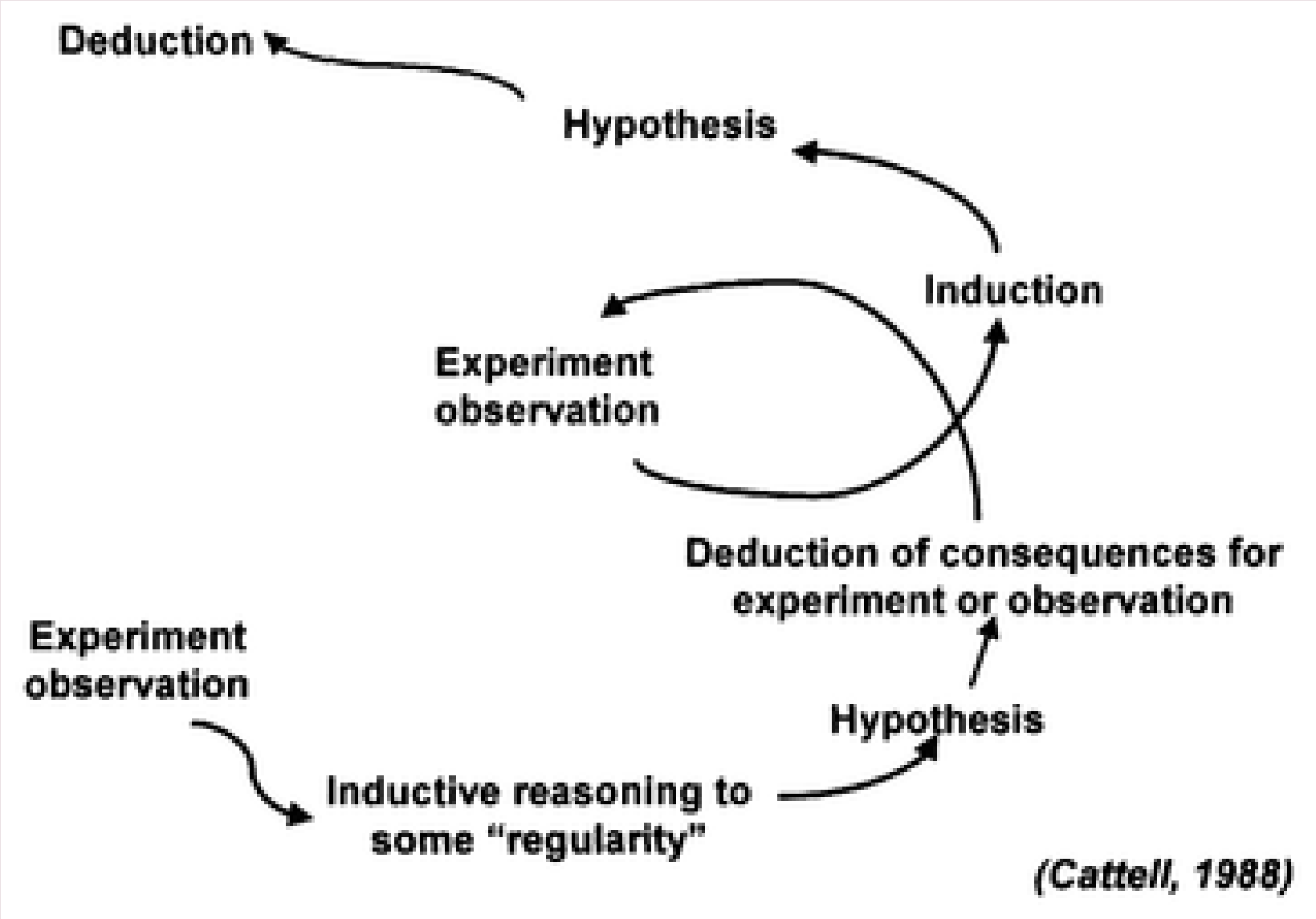
Results/Findings
Analysis/Interpretation

Conclusions

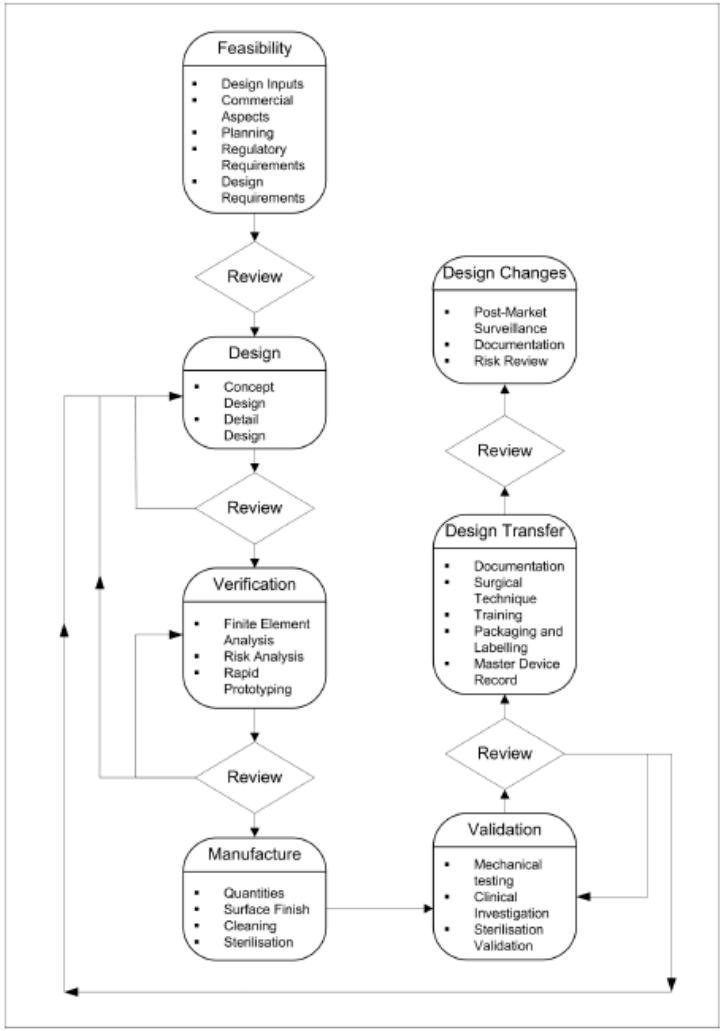
See Flynn & Ingerson (2021)
Chapter 1: Understanding Research And Ethics In *Research Design for the Behavioral Sciences An Applied Approach* (p.13-15) https://connect.springerpub.com/content/book/978-0-8261-4385-3/part/part01/chapter/ch01

Inductive and Deductive Reasoning




Deduction
Hypothesis
Induction
Experiment
observation
Deduction of consequences for
experiment or observation
Experiment
observation
Hypothesis
Inductive reasoning to
some "regularity"
(Cattell, 1988)

A Review of the Design Process for Implantable Orthopedic Medical Devices Fig 1 p.22
Article in The Open Biomedical Engineering Journal · Aitchison & Hukins February 2009



Sections and Structure

4. Discussion – recommendations and future work

Sections and Structure

5. References

Sections and Structure

6. Appendices

CW5 Mark Sheet:
MSc Project Final Report (Extended Research Project).

Criteria		Assessment Criteria CW5	Marks Range	Mark Awarded
1	BACKGROUND, INTRODUCTION and LITERATURE REVIEW - Quality of coverage and context (__/20)	[Clear Fail]: Essentially no background material presented or few www references.	0-7	
		[Borderline Fail]: Scientific background poorly conveyed; absent key references/concepts.	8-9	
		[Pass]: Most scientific aims identified; some confusion relating to prioritisation and relevance.	10-11	
		[Merit]: Coherent narrative; almost all aims outlined.	12-13	
		[Exceptional]: Clear display of subject mastery; precise, coherent, structured; all key refs. cited	17-20	
2	UNDERSTANDING & ANALYSIS - Scientific awareness, justification (__/30)	[Clear Fail]: Demonstrates essentially no understanding of project; no meaningful analysis.	0-11	
		[Borderline Fail]: Very limited grasp of key scientific issues; little evidence of critical analysis.	12-14	
		[Pass]: Some comprehension of scientific issues; attempts to apply reason-based analysis.	15-17	
		[Merit]: Appreciation of scientific challenges; reasoned justification of strategic decisions taken	18-20	
		[Distinction]: Clear awareness of scientific challenges; logical approach to problem solving	21-25	
		[Exceptional]: Lucid presentation of nub of challenges faced; evidence-based reasoning and analysis.	26-30	
3	QUALITY OF RESULTS & DISCUSSION - Extent and accuracy (__/30)	[Clear Fail]: Quality of data doubtful; format unconventional	0-11	
		[Borderline Fail]: Presentation of data imprecise or inaccurate	12-14	
		[Pass]: Some useful standard data presented; format and presentation non-ideal	15-17	
		[Merit]: Useful data presented (or papers discussed); most important details documented; some inconsistencies	18-20	
		[Distinction]: High quality results and discussion of publication standard (data from multiple papers compared); format largely adheres to convention	21-25	
		[Exceptional]: Outstanding high-quality results and discussion; substantial project progress mad	26-30	
4	PRESENTATION - Structure, clarity, written style, quality of English (__/20)	[Clear Fail]: Lacking any apparent order; English usage very poor; no links between text and figures/tables	0-7	
		[Borderline Fail]: Minimal organisation of material; difficult to follow; unclear; poor English	8-9	
		[Pass]: Apparent structure into sections; English acceptable	10-11	
		[Merit]: English good; organisation of material into logical sections; clear style	12-13	
		[Distinction]: Essentially error free; clear logical construction; balanced presentation of message	14-16	
		[Exceptional]: Flawless English; clear, logical structure; engaging style; well-developed message	17-20	
			Total	

CW5: *Project Final Report (Extended Research Project)*
AHEP4 M2: *Generate a final project report to build on complex scientific/engineering principles applied to the research project to show how an analysis of project outcomes support project conclusions.*
AHEP4 M3: *Select and apply appropriate computational and analytical techniques to model complex problems, recognising the limitations of the techniques employed.*
AHEP4 M4: *Update and develop technical literature reviews and other sources of information to inform project discussions relating to complex engineering/scientific problems beyond the limitations and constraints of the final project report.*
AHEP4 M5: *Provide a final report that contains design solutions for complex problems that meet a combination of societal, user, business and customer needs, as appropriate.*
AHEP4 M16*: Function effectively as an individual and as a member of team working with academics, project supervisors and laboratory/workshop technical staff. Use the extended engineering/scientific literature review to develop and communicate the rationale, aims and objectives (RAO) to project supervisors and academics.*
AHEP4 M17: *Generate a Final Report to help communicate effectively on complex engineering matters to technical and non-technical audiences to show fulfilment of project rationale, aims and objectives (RAO).*

Planning Activity:
Final Report Structure

MSc Project Viva

- An oral examination.
- Pre-prepared project summary presentation for the viva examiner. The project presentation needs to be 3 minutes (maximum) and kept concise by clearly covering, for example, project background & situation/problem, project justification, project RAO, project outcomes, findings, implications & conclusions.
- You will be asked by your examiners about your project e.g., the rationale, your approach, the data, the results, analysis, findings, implications and conclusions.
- You need to be prepared and anticipate questions about the work in your final report.
- Be prepared to justify and debate sections in the report - demonstrate your knowledge, understanding, purposefulness, relevance, usefulness and benefits from carrying out the work.
- Be clear about the original, novel and/or innovative contribution you have made through your project.

CW6 Mark Sheet:
MSc Project Viva (Oral Examination).

Criteria		Assessment Criteria CW6	Marks Range	Mark Awarded
1	SUMMARY and PRESENTING the PROJECT __/50	<25 [Fail]	0-24	
		25-29 [Pass]	25-29	
		30-34 [Merit]	30-34	
		35-40 [Distinction]	35-40	
		41-50 [Exceptional]	41-50	
2	VIVA DISCUSSION - Extent, depth and accuracy __/50	<25 [Fail]	0-24	
		25-29 [Pass]	25-29	
		30-34 [Merit]	30-34	
		35-40 [Distinction]	35-40	
		41-50 [Exceptional]	41-50	
			Total	

CW6: *Oral examination – Viva*
AHEP4 M16*: Participate in an oral presentation and review of the project effectively as an individual, and as a member of the project supervision team.*
AHEP4 M17: *Under Viva conditions, communicate effectively on complex engineering matters with technical and non-technical audiences. Explain and justify project via a presentation and through responding respond to questions and queries that challenge the project rationale, approach, methodology, results/outcomes and conclusions.*

1. Viva will be held with both an **examiner (marker 2) and your supervisor (marker 1)**.
2. The examiner will lead the viva, but both your examiner and supervisor will separately assess your viva (oral examination).
3. It is recommended that you **prepare a concise 2-3 minute presentation** (**but 3 minutes maximum**) to summarise your extended research project (e.g., rationale, aim, objectives, specific problem(s) or context, method/approach, justification, outcomes (results, findings and outcomes). A concise summary of your extended research purpose and outcomes will help you prepare for your viva and it will also give you an opportunity to cover the salient points from from your project report for your examiner at the start of your oral examination (viva). PowerPoint can be used, but you can use other formats for your project summary. **Remember to upload a pdf file of your viva presentation to both viva submission points before your viva.**
4. Remember that your project summary needs to be short, clear and concise e.g., what you did; how and why you did it; what was the outcome (results & conclusion); why was it useful and how did you address/meet the project RAO.
5. The **oral examination (viva) will take 15-20 minutes**, and during this time you will be asked questions about your project and the final report. Your examiner and supervisor **will already have received your final report so it is recommended that also you have your report to hand during the viva** so that, if necessary, you can refer to sections of the report to help support and justify your answers, explanations and reasoning.
6. The viva is an **opportunity to show the work you did and demonstrate your knowledge, understanding, analysis** driven by clear purpose and rationale.
7. The **viva assessment criteria** will be based on the following areas (see below).